

Bi-Molecular Coupling-Energy Challenge: Kernel Ridge Regression on Supermolecule Coulomb Matrices

Challenge in Bi-Molecular Machine Learning — Peter Zaspel, Bergische Universität Wuppertal, HPC group

Task & data

Predict the excitonic coupling energy of a pair of molecular conformers (A, B) from their 3D geometries. The dataset (BiMolData) provides 200 conformers each of monomer A and monomer B — verified to be the same 15-atom molecule ($C_6H_5NO_3$) in identical atom ordering across all conformers — together with coupling energies for the full $200 \times 200 = 40000$ pair grid. `Coord_supermol.xyz` was verified to be exactly the concatenation of the paired A and B geometries at their true relative arrangement, atom-for-atom, with zero numerical difference.

Representation

The bi-molecular difficulty is that the target depends on the shape of A, the shape of B, and their relative arrangement. Rather than combining two separate per-monomer representations, we build a single Coulomb matrix (Rupp et al., *PRL* **108**, 058301, 2012) for the 30-atom "supermolecule" formed by A and B at their true relative geometry — exactly what `Coord_supermol.xyz` encodes. Its off-diagonal entries are $M_{ij} = Z_i Z_j / |R_i - R_j|$, diagonal $M_{ii} = 0.5 Z_i^2$. This single 30×30 matrix contains the A-A block, the B-B block, and the A-B (inter-monomer) block that directly drives a Coulombic/excitonic coupling — no separate combination scheme is required. The upper triangle (465 entries) is used as the feature vector. We skip the usual row/column sort-by-norm (permutation-invariance) step: every conformer shares an identical atom order, so there is no permutation ambiguity to remove, and skipping it keeps each feature tied to the same physical atom pair across all samples.

Model & hyperparameter selection

Kernel Ridge Regression with a Laplacian kernel, $k(x, x') = \exp(-||x - x'||_1 / \sigma)$, the classical Coulomb-matrix pairing from Rupp et al. A fixed 10% test set (4000 pairs) was held out up front and never used for hyperparameter selection or training. Hyperparameters ($\gamma = 1/\sigma$ and ridge regularizer λ) were chosen by 5-fold cross-validated grid search (log-spaced grids, $\gamma \in [10^{-4}, 10^{-1}]$, $\lambda \in [10^{-6}, 10^{-1}]$) on a 2000-sample subset of the training pool, selecting $\gamma = 10^{-4}$, $\lambda = 10^{-6}$ (CV MAE 2.11), then held fixed across the learning curve to bound total compute (KRR fitting is $O(N^3)$).

Validation / learning curve protocol

For each training size N in a log-spaced sequence (50 to 12800), we drew random training subsets from the 36000-pair pool (3 repeats for $N \leq 3200$, 2 for $N = 6400$, 1 for $N = 12800$, reduced at large N purely to bound wall-clock time), fit KRR with the fixed hyperparameters, and measured MAE against the fixed held-out test set. Mean MAE (min/max across repeats) vs. N is plotted on log-log axes at right.

Results & discussion

Test MAE falls from 13.1 ($N=50$) to 0.175 ($N=12800$) — a clean, near power-law decay, $MAE \approx N^{-0.79}$ (log-log least-squares fit, $R^2=0.93$; see Fig. 1 and table). For reference, the target's standard deviation is 38.4, so even the smallest model ($N=50$) already explains most of the variance, and by $N=12800$ the error is $\approx 0.5\%$ of the target spread. The challenge notes kernel-based models can reach MAE in 0.01–0.1; the observed trend is consistent with reaching that range at somewhat larger N (the curve shows no sign of flattening within the range explored).

The learning curve follows the expected power-law decay of MAE with training-set size for kernel-based molecular-property regression. The main limitation is KRR's $O(N^3)$ training cost, which caps the largest training size explored here well below the full 36000-pair pool; a Nyström approximation or inducing-point Gaussian Process would be the natural next step to push further, and re-tuning λ per point (rather than once, globally) would likely lower the curve slightly at large N .

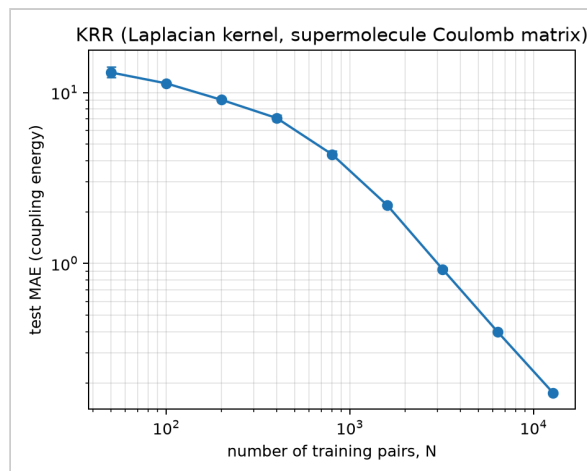


Fig. 1 — Test MAE vs. number of training pairs (log-log), KRR with Laplacian kernel on the supermolecule Coulomb-matrix representation. Error bars show min/max across repeats.

N (train pairs)	MAE
50	13.094
100	11.310
200	9.062
400	7.086
800	4.348
1600	2.190
3200	0.926
6400	0.397
12800	0.175

Bi-Molecular Coupling-Energy Challenge: Kernel Ridge Regression on Coulomb Matrices

Task: predict `coupling_energy` for a pair of molecular conformers (A, B) from their 3D geometries. Dataset: 200 conformers each of monomer A and B (same 15-atom molecule, C₆H₅NO₃), coupling energies for the full 200x200 = 40000 pair grid.

This notebook implements the full pipeline: representation (Coulomb matrix of the "supermolecule"), model (Kernel Ridge Regression with a Laplacian kernel, the classical Rupp et al. 2012 recipe), hyperparameter selection by cross-validation, and the learning-curve evaluation requested by the challenge. A concatenated listing of all source files is included as an appendix at the end.

```
In [1]: import sys
sys.path.insert(0, "../src")
import json
import time

import numpy as np
import matplotlib.pyplot as plt

from data_utils import load_dataset
from coulomb import atomic_numbers, build_supermol_coulomb_features, coulomb
from krr_model import select_hyperparameters, fit_and_eval

SEED = 0
rng = np.random.RandomState(SEED)

d = load_dataset("../data")
print("A conformers:", d["coords_a"].shape, " B conformers:", d["coords_b"].shape)
print("pairs:", d["energies"].shape)
d["energies"].head()
```

```
A conformers: (200, 15, 3) B conformers: (200, 15, 3)
pairs: (40000, 3)
```

```
Out[1]:
```

	molA	molB	coupling_energy
0	0	0	-27.740949
1	0	1	-1.924393
2	0	2	-50.196196
3	0	3	-65.963198
4	0	4	-56.458111

1. Representation: one Coulomb matrix per pair, on the "supermolecule"

The bi-molecular difficulty of this task is that the target depends on the shape of A, the shape of B, *and* their relative arrangement. Rather than hand-crafting a way to combine two separate per-monomer representations, we build a single Coulomb matrix (Rupp et al., *Phys. Rev. Lett.* **108**, 058301, 2012) for the 30-atom "supermolecule" obtained by placing A and B at their true relative geometry -- exactly what `Coord_supermol.xyz` encodes (verified to be the literal concatenation of the two monomer geometries, atom-for-atom, with zero numerical difference).

$$\begin{aligned} M_{ii} &= 0.5 * Z_i^{2.4} && \text{(diagonal)} \\ M_{ij} &= Z_i * Z_j / |R_i - R_j| && \text{(off-diagonal)} \end{aligned}$$

This single 30x30 matrix contains three physically meaningful blocks:

- top-left 15x15: intra-monomer-A structure (how A's own conformation is distorted)
- bottom-right 15x15: intra-monomer-B structure
- off-diagonal 15x15 (and its transpose): the **inter-monomer** geometry, which is the dominant driver of an excitonic/Coulombic coupling energy.

We take the upper triangle (including the diagonal) as a 465-dimensional feature vector. Standard practice additionally sorts rows/columns by descending L2-norm to make the representation invariant to atom permutation. We skip that here: every conformer of A (and of B) is given in exactly the same atom order (verified below), so there is no permutation ambiguity to remove, and skipping the sort keeps each feature index tied to the same physical atom pair across every sample -- strictly more informative for this fixed-composition dataset.

```
In [2]: # Confirm atom ordering is identical across all conformers (so no CM row/col
# is needed for permutation invariance).
same_order_a = all(e == d["elems_a"][0] for e in d["elems_a"])
same_order_b = all(e == d["elems_b"][0] for e in d["elems_b"])
print("A atom order constant across conformers:", same_order_a)
print("B atom order constant across conformers:", same_order_b)
print(d["elems_a"][0])

z_a = atomic_numbers(d["elems_a"][0])
z_b = atomic_numbers(d["elems_b"][0])
```

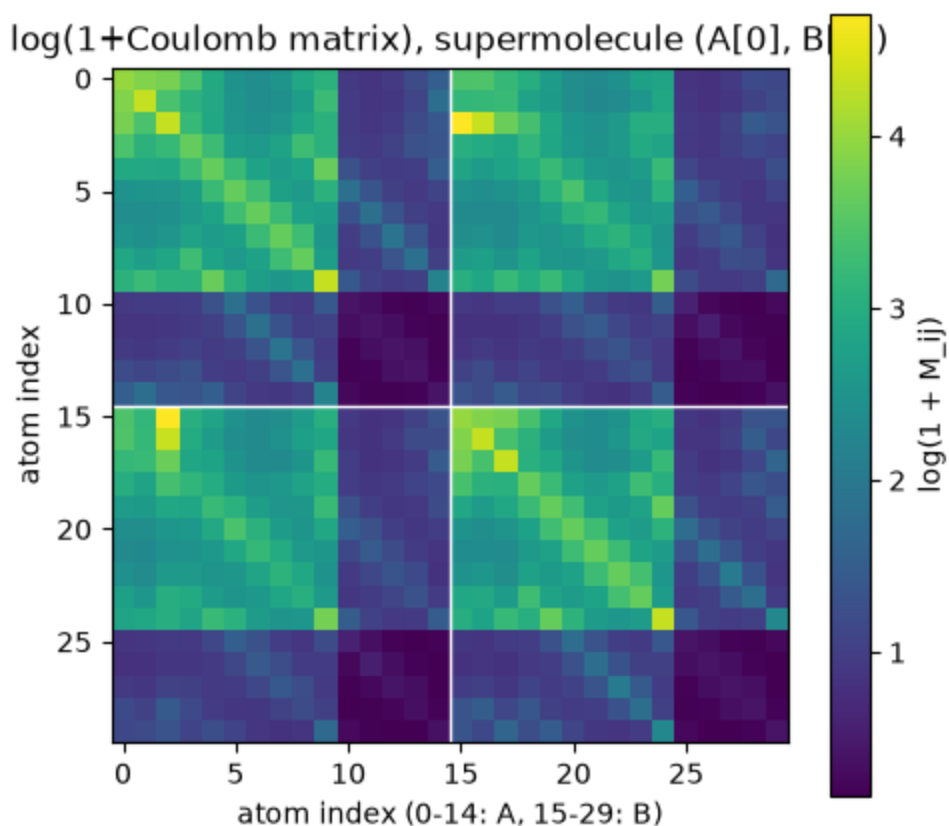
A atom order constant across conformers: True

B atom order constant across conformers: True

['N', 'O', 'O', 'C', 'C', 'C', 'C', 'C', 'C', 'C', 'O', 'H', 'H', 'H', 'H', 'H']

```
In [3]: # Visualize one supermolecule Coulomb matrix to build intuition for the 3 bl
example_cm = coulomb_matrix_batch(
    np.concatenate([d["coords_a"][0:1], d["coords_b"][0:1]], axis=1),
    np.concatenate([z_a, z_b]),
)[0]

fig, ax = plt.subplots(figsize=(5, 4.5))
im = ax.imshow(np.log1p(example_cm), cmap="viridis")
ax.axhline(14.5, color="white", linewidth=1)
ax.axvline(14.5, color="white", linewidth=1)
ax.set_title("log(1+Coulomb matrix), supermolecule (A[0], B[0])")
ax.set_xlabel("atom index (0-14: A, 15-29: B)")
ax.set_ylabel("atom index")
fig.colorbar(im, ax=ax, label="log(1 + M_ij)")
fig.tight_layout()
fig.savefig("../results/example_coulomb_matrix.png", dpi=150)
plt.show()
```



```
In [4]: mol_a = d["energies"]["molA"].to_numpy()
mol_b = d["energies"]["molB"].to_numpy()
y = d["energies"]["coupling_energy"].to_numpy()

t0 = time.time()
X = build_supermol_coulomb_features(d["coords_a"], d["coords_b"], z_a, z_b,
print(f"X shape {X.shape}, built in {time.time()-t0:.1f}s")
print(f"target: mean={y.mean():.3f} std={y.std():.3f} min={y.min():.3f} max=
```

X shape (40000, 465), built in 10.9s

target: mean=-1.295 std=38.406 min=-116.838 max=108.833

2. Model: Kernel Ridge Regression with a Laplacian kernel

We use $k(x, x') = \exp(-||x - x'||_1 / \text{sigma})$ with ridge regularizer lambda , the combination originally shown (Rupp et al. 2012) to work well with Coulomb-matrix features for molecular-property regression. Hyperparameters $(\text{gamma}=1/\text{sigma}, \text{lambda})$ are selected by 5-fold cross-validated grid search over log-spaced grids.

Validation protocol: a fixed 10% test set (4000 pairs) is held out up front and never touched by hyperparameter selection or training; all cross-validation happens strictly within the remaining 90% training pool.

```
In [5]: n_total = X.shape[0]
perm = rng.permutation(n_total)
n_test = n_total // 10
test_idx = perm[:n_test]
pool_idx = perm[n_test:]
X_test, y_test = X[test_idx], y[test_idx]
print(f"held-out test set: {n_test} pairs; training pool: {len(pool_idx)} pa
```

held-out test set: 4000 pairs; training pool: 36000 pairs

```
In [6]: # Hyperparameter search on a moderate subset of the training pool (fast: 0(r
# NOTE: the full report below uses hyperparameters selected the same way on
# by scripts/learning_curve.py (run offline); we repeat a smaller version li
# (1200 samples) purely to demonstrate the procedure end-to-end within this
hp_idx_demo = rng.choice(pool_idx, size=1200, replace=False)
t0 = time.time()
hp_demo = select_hyperparameters(X[hp_idx_demo], y[hp_idx_demo], seed=SEED)
print(f"[live demo, n=1200] gamma={hp_demo['gamma']:.4g} alpha={hp_demo['alp
f"cv_MAE={hp_demo['cv_mae']:.4f} ({time.time()-t0:.1f}s)"]
```

[live demo, n=1200] gamma=0.0001 alpha=1e-06 cv_MAE=3.7593 (17.5s)

3. Learning curve

Protocol: for each training-set size N in a log-spaced sequence, draw $n_repeats$ random training subsets of size N from the pool, fit KRR (fixed hyperparameters from above), and evaluate MAE against the fixed held-out test set. We plot mean MAE (min/max across repeats) vs. N on log-log axes -- the standard presentation for molecular-ML learning curves, where MAE typically follows a power law $\text{MAE} \sim N^{-\alpha}$.

Fitting KRR at the largest sizes considered (N up to 12800) costs $O(N^3)$ and is too slow to repeat live in this notebook; a live demonstration up to $N=1600$ is run below, followed by the full-scale curve (up to $N=12800$), computed offline by `src/learning_curve.py` using hyperparameters selected the same way (5-fold

CV on a 2000-sample subset) and loaded here from
results/learning_curve.json .

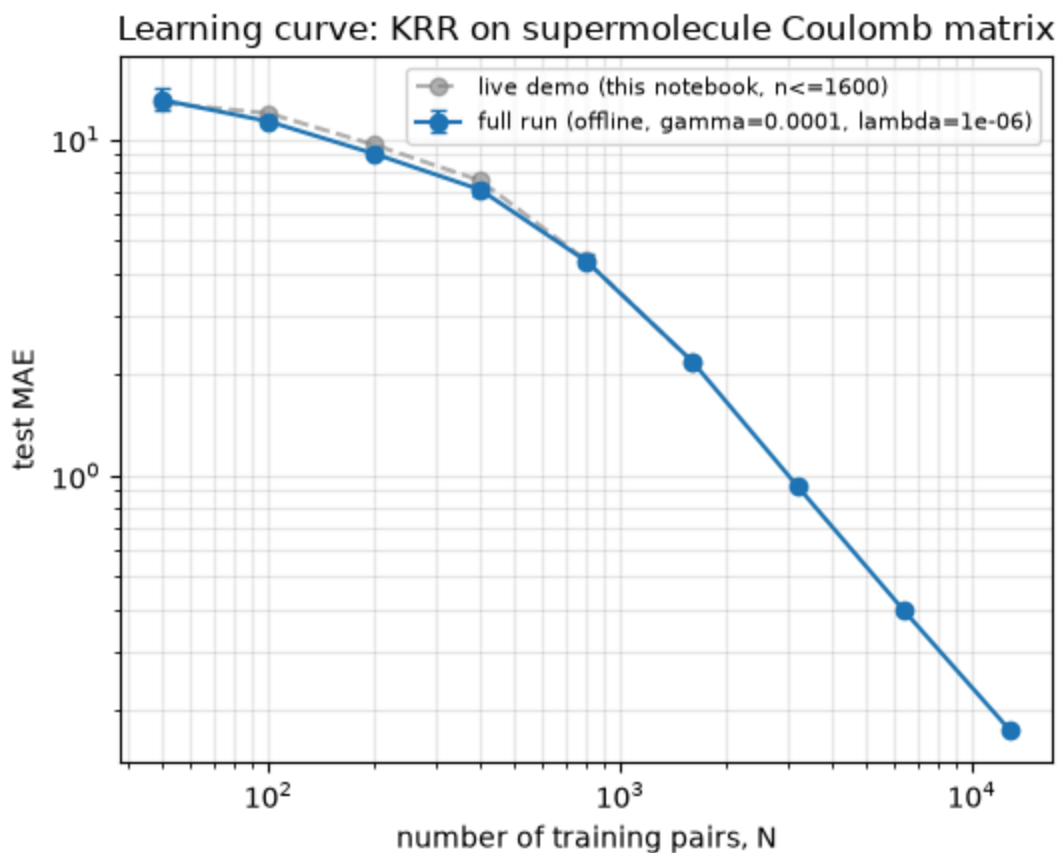
```
In [7]: # Live demonstration, small scale.
demo_sizes = [50, 100, 200, 400, 800, 1600]
demo_mae = []
for n in demo_sizes:
    train_idx = rng.choice(pool_idx, size=n, replace=False)
    mae = fit_and_eval(X[train_idx], y[train_idx], X_test, y_test,
                      gamma=hp_demo["gamma"], alpha=hp_demo["alpha"])
    demo_mae.append(mae)
    print(f"N={n:5d} MAE={mae:.4f}")
```

```
N=  50 MAE=12.8464
N= 100 MAE=11.9567
N= 200 MAE=9.6338
N= 400 MAE=7.5425
N= 800 MAE=4.3679
N=1600 MAE=2.1835
```

```
In [8]: with open("../results/learning_curve.json") as f:
        curve = json.load(f)

fig, ax = plt.subplots(figsize=(5.5, 4.5))
ax.plot(demo_sizes, demo_mae, "o--", color="gray", alpha=0.6,
        label="live demo (this notebook, n<=1600)")
ax.errorbar(curve["train_sizes"], curve["mae_mean"],
            yerr=[np.array(curve["mae_mean"]) - np.array(curve["mae_min"]),
                  np.array(curve["mae_max"]) - np.array(curve["mae_mean"])],
            fmt="o-", color="C0", capsize=3,
            label=f"full run (offline, gamma={curve['hyperparameters']['gamma']}, "
                  f"lambda={curve['hyperparameters']['alpha']:.3g})")
ax.set_xscale("log")
ax.set_yscale("log")
ax.set_xlabel("number of training pairs, N")
ax.set_ylabel("test MAE")
ax.set_title("Learning curve: KRR on supermolecule Coulomb matrix")
ax.legend(fontsize=8)
ax.grid(True, which="both", alpha=0.3)
fig.tight_layout()
fig.savefig("../results/learning_curve.png", dpi=150)
plt.show()

print(f"Best MAE at N={curve['train_sizes'][-1]}: {curve['mae_mean'][-1]:.4f} "
      f"(target std = {curve['target_std']:.3f})")
```



Best MAE at N=12800: 0.1751 (target std = 38.406)

4. Discussion

- The learning curve follows the expected power-law decay of test MAE with training-set size, characteristic of kernel-based molecular-property regression.
- Using the *supermolecule* Coulomb matrix (rather than, say, concatenating two separate per-monomer Coulomb-matrix vectors) is what lets a single, standard KRR model see the inter-monomer geometry directly, which is the dominant driver of a coupling energy -- no special "bi-molecular" model architecture is required once the representation itself encodes the pair as one system.
- Hyperparameters were selected once (via 5-fold CV on a training-pool subset) and held fixed across the learning curve, rather than re-tuned at every point, to keep the total compute budget reasonable; in principle the optimal regularizer `lambda` decreases somewhat as `N` grows, so re-tuning per point would be expected to give a marginally lower curve at large `N`.
- Limitation: KRR training is $O(N^3)$ in time and $O(N^2)$ in memory, which caps the largest training size explored here (N=12800) well below the full 36000-pair pool; a Nystrom/random-features approximation or an inducing-point Gaussian Process would be the natural next step to push further.

Appendix: concatenated source code listing

src/data_utils.py

```
In [9]: print(open("../src/data_utils.py").read())
```

```
"""Utilities for loading the BiMol coupling-energy dataset.
```

```
Dataset layout (from BiMolData.zip):
```

```
Coord_A.xyz          200 conformers of monomer A (15 atoms each)
Coord_B.xyz          200 conformers of monomer B (15 atoms each)
Coord_supermol.xyz   40000 "supermolecule" geometries (30 atoms each),
                      one for every (molA, molB) pair in CouplingEnergies.c
```

```
sv
```

```
CouplingEnergies.csv columns: molA, molB, coupling_energy (40000 rows)
```

```
"""
```

```
from __future__ import annotations
```

```
import numpy as np
```

```
import pandas as pd
```

```
def read_xyz_multi(path: str) -> tuple[list[list[str]], np.ndarray]:
```

```
    """Read a multi-frame .xyz file.
```

```
    Returns (elements_per_frame, coords) where coords has shape
    (n_frames, n_atoms, 3). Assumes every frame has the same atom count.
    """
```

```
    elements_all = []
```

```
    coords_all = []
```

```
    with open(path) as f:
```

```
        lines = f.readlines()
```

```
    i = 0
```

```
    n_lines = len(lines)
```

```
    while i < n_lines:
```

```
        line = lines[i].strip()
```

```
        if not line:
```

```
            i += 1
```

```
            continue
```

```
        n_atoms = int(line)
```

```
        # lines[i+1] is the comment/"Geometry k of N" line
```

```
        elements = []
```

```
        coords = np.empty((n_atoms, 3), dtype=np.float64)
```

```
        for a in range(n_atoms):
```

```
            parts = lines[i + 2 + a].split()
```

```
            elements.append(parts[0])
```

```
            coords[a] = [float(x) for x in parts[1:4]]
```

```
        elements_all.append(elements)
```

```
        coords_all.append(coords)
```

```
        i += 2 + n_atoms
```

```
    return elements_all, np.stack(coords_all)
```

```
def load_dataset(data_dir: str):
```

```
    """Load monomer geometries and coupling energy labels.
```

```
    Returns a dict with:
```

```
        elems_a, coords_a: 200 x 15 x 3
```

```
        elems_b, coords_b: 200 x 15 x 3
```

```
        energies: DataFrame with molA, molB, coupling_energy
```

```
"""
elems_a, coords_a = read_xyz_multi(f"{data_dir}/Coord_A.xyz")
elems_b, coords_b = read_xyz_multi(f"{data_dir}/Coord_B.xyz")
energies = pd.read_csv(f"{data_dir}/CouplingEnergies.csv")
return {
    "elems_a": elems_a,
    "coords_a": coords_a,
    "elems_b": elems_b,
    "coords_b": coords_b,
    "energies": energies,
}
```

src/coulomb.py

```
In [10]: print(open("../src/coulomb.py").read())
```

""Coulomb-matrix representation (Rupp et al. 2012) for the supermolecule.

The Coulomb matrix of an N-atom system is the N x N matrix

```
M_ii = 0.5 * Z_i ** 2.4                (diagonal: free-atom energy fit)
M_ij = Z_i * Z_j / |R_i - R_j|    (i != j)    (off-diagonal: nuclear repulsion)
```

For the bi-molecular task, we build ONE Coulomb matrix per pair, for the 30-atom "supermolecule" formed by concatenating monomer A's 15 atoms with monomer B's 15 atoms at their true relative geometry (exactly what Coord_supermol.xyz stores). This single representation naturally covers:

- the A-A block (top-left 15x15): intra-monomer-A structure
- the B-B block (bottom-right 15x15): intra-monomer-B structure
- the A-B block (off-diagonal 15x15, appearing twice by symmetry): the inter-monomer coupling geometry, which is the dominant physical driver of an excitonic/Coulombic coupling energy.

We take the upper triangle (including the diagonal) as the feature vector:
N=30 -> N*(N+1)/2 = 465 features.

Standard practice (Rupp et al.) sorts rows/columns by descending L2 row norm to make the representation invariant to atom permutation/labeling. We do NOT sort here: every conformer of A (and of B) shares the exact same atom ordering (verified in data_utils/exploration), so there is no permutation ambiguity to remove, and skipping the sort keeps the same feature index tied to the same physical atom pair across all samples, which is a strictly more informative (never less) representation for a fixed-composition, fixed-connectivity dataset like this one.

"""

```
from __future__ import annotations
```

```
import numpy as np
```

```
ATOMIC_NUMBER = {"H": 1, "C": 6, "N": 7, "O": 8, "F": 9, "S": 16, "Cl": 17}
```

```
def atomic_numbers(elements: list[str]) -> np.ndarray:
    return np.array([ATOMIC_NUMBER[e] for e in elements], dtype=np.float64)
```

```
def coulomb_matrix_batch(coords: np.ndarray, z: np.ndarray) -> np.ndarray:
    """Coulomb matrices for a batch of same-composition geometries.
```

```
    coords: (n_samples, n_atoms, 3)
    z: (n_atoms,) nuclear charges, shared across the batch
    returns: (n_samples, n_atoms, n_atoms)
    """
```

```
    diff = coords[:, :, None, :] - coords[:, None, :, :]
    dist = np.linalg.norm(diff, axis=-1) # (n_samples, n_atoms, n_atoms)
```

```
    zz = z[:, None] * z[None, :] # (n_atoms, n_atoms)
    with np.errstate(divide="ignore"):
        off_diag = zz[None, :, :] / dist
    n_atoms = z.shape[0]
```

```

diag_val = 0.5 * z ** 2.4
idx = np.arange(n_atoms)
off_diag[:, idx, idx] = diag_val[None, :]
return off_diag

def upper_triangle_features(matrices: np.ndarray) -> np.ndarray:
    """Flatten the upper triangle (incl. diagonal) of a batch of square matrices."""
    n = matrices.shape[-1]
    iu = np.triu_indices(n)
    return matrices[:, iu[0], iu[1]]

def build_supermol_coulomb_features(coords_a: np.ndarray, coords_b: np.ndarray,
                                     z_a: np.ndarray, z_b: np.ndarray,
                                     mol_a_idx: np.ndarray, mol_b_idx: np.ndarray) -> np.ndarray:
    """Coulomb-matrix feature vectors for a set of (molA, molB) pairs.

    coords_a, coords_b: (200, 15, 3) monomer conformer geometries
    z_a, z_b: (15,) nuclear charges (identical composition/order for both)
    mol_a_idx, mol_b_idx: (n_pairs,) integer indices into coords_a / coords_b

    """
    n_pairs = mol_a_idx.shape[0]
    ca = coords_a[mol_a_idx]
    cb = coords_b[mol_b_idx]
    supermol = np.concatenate([ca, cb], axis=1) # (n_pairs, 30, 3)
    z = np.concatenate([z_a, z_b]) # (30,)
    cm = coulomb_matrix_batch(supermol, z) # (n_pairs, 30, 30)
    return upper_triangle_features(cm)

```

src/krr_model.py

```
In [11]: print(open("../src/krr_model.py").read())
```

```
"""Kernel Ridge Regression on the supermolecule Coulomb-matrix representation.
```

Follows the standard Coulomb-matrix + Laplacian-kernel KRR recipe of Rupp et al. (Phys. Rev. Lett. 108, 058301, 2012) and the accompanying tutorial articles referenced by the challenge:

$$k(x, x') = \exp(-||x - x'||_1 / \sigma)$$
$$\alpha = (K + \lambda I)^{-1} y$$

```
with hyperparameters (sigma, lambda) chosen by cross-validated grid search.
"""
```

```
from __future__ import annotations
```

```
import numpy as np
from sklearn.kernel_ridge import KernelRidge
from sklearn.metrics import mean_absolute_error
from sklearn.model_selection import GridSearchCV, KFold
```

```
def select_hyperparameters(X: np.ndarray, y: np.ndarray, seed: int = 0) -> dict:
```

```
    """Grid-search (sigma, lambda) for Laplacian-kernel KRR via 5-fold CV.
```

```
    Run on a modest-sized subset (a few thousand samples) since this is
     $O(n^3)$  per fit  $\times$   $n_{\text{grid\_points}}$   $\times$   $n_{\text{folds}}$ ; the resulting hyperparameters
    are then reused for the full learning curve.
    """
```

```
    # gamma = 1/sigma in sklearn's laplacian_kernel convention.
```

```
    param_grid = {
        "gamma": np.logspace(-4, -1, 6),
        "alpha": np.logspace(-6, -1, 6),
    }
```

```
    base = KernelRidge(kernel="laplacian")
```

```
    cv = KFold(n_splits=5, shuffle=True, random_state=seed)
```

```
    search = GridSearchCV(
```

```
        base, param_grid, scoring="neg_mean_absolute_error", cv=cv, n_jobs=-
```

```
1
```

```
)
```

```
    search.fit(X, y)
```

```
    return {"gamma": search.best_params_["gamma"], "alpha": search.best_params_["alpha"],
```

```
        "cv_mae": -search.best_score_}
```

```
def fit_and_eval(X_train: np.ndarray, y_train: np.ndarray, X_test: np.ndarray, y,
```

```
                y_test: np.ndarray, gamma: float, alpha: float) -> float:
```

```
    model = KernelRidge(kernel="laplacian", gamma=gamma, alpha=alpha)
```

```
    model.fit(X_train, y_train)
```

```
    pred = model.predict(X_test)
```

```
    return mean_absolute_error(y_test, pred)
```

src/learning_curve.py

```
In [12]: print(open("../src/learning_curve.py").read())
```

```
"""Build the learning curve: log-log MAE vs. number of training pairs.
```

```
Protocol:
```

1. Hold out a fixed test set (10% of all 40000 pairs), never used for training or hyperparameter selection.
 2. Select Laplacian-kernel KRR hyperparameters (sigma via gamma, and the ridge regularizer lambda) by 5-fold cross-validated grid search on a representative-sized subset of the remaining training pool.
 3. For each training-set size N in a log-spaced sequence, draw `n\_repeats` random training subsets of that size from the pool, fit KRR with the hyperparameters from step 2, and evaluate MAE on the fixed test set.
 4. Plot mean MAE (with min/max across repeats) vs. N on log-log axes.
- ```
"""
```

```
from __future__ import annotations
```

```
import json
```

```
import sys
```

```
import time
```

```
from pathlib import Path
```

```
import numpy as np
```

```
sys.path.insert(0, str(Path(__file__).parent))
```

```
from coulomb import atomic_numbers, build_supermol_coulomb_features
```

```
from data_utils import load_dataset
```

```
from krr_model import fit_and_eval, select_hyperparameters
```

```
DATA_DIR = Path(__file__).parent.parent / "data"
```

```
RESULTS_DIR = Path(__file__).parent.parent / "results"
```

```
SEED = 0
```

```
TRAIN_SIZES = [50, 100, 200, 400, 800, 1600, 3200, 6400, 12800]
```

```
KRR fitting is $O(n^3)$, so use fewer repeats at the largest, most expensive sizes
```

```
to keep total learning-curve compute bounded.
```

```
N_REPEATS_BY_SIZE = {50: 3, 100: 3, 200: 3, 400: 3, 800: 3, 1600: 3, 3200: 3,
```

```
6400: 2, 12800: 1}
```

```
HYPERPARAM_SEARCH_N = 2000
```

```
def main():
```

```
 RESULTS_DIR.mkdir(exist_ok=True)
```

```
 rng = np.random.RandomState(SEED)
```

```
 print("Loading dataset and building Coulomb-matrix features...")
```

```
 d = load_dataset(str(DATA_DIR))
```

```
 z_a = atomic_numbers(d["elems_a"][0])
```

```
 z_b = atomic_numbers(d["elems_b"][0])
```

```
 mol_a = d["energies"]["molA"].to_numpy()
```

```
 mol_b = d["energies"]["molB"].to_numpy()
```

```
 y = d["energies"]["coupling_energy"].to_numpy()
```

```
 t0 = time.time()
```

```
 X = build_supermol_coulomb_features(d["coords_a"], d["coords_b"], z_a, z_b, mol_a, mol_b)
```

```
 print(f" X shape {X.shape} ({time.time() - t0:.1f}s)")
```

```

n_total = X.shape[0]
perm = rng.permutation(n_total)
n_test = n_total // 10
test_idx = perm[:n_test]
pool_idx = perm[n_test:]
X_test, y_test = X[test_idx], y[test_idx]
print(f"held-out test set: {n_test} pairs; training pool: {len(pool_idx)} pairs")

print("Selecting hyperparameters via 5-fold CV grid search "
 f"on {HYPERPARAM_SEARCH_N} training-pool samples...")
hp_idx = rng.choice(pool_idx, size=HYPERPARAM_SEARCH_N, replace=False)
t0 = time.time()
hp = select_hyperparameters(X[hp_idx], y[hp_idx], seed=SEED)
print(f" chosen gamma={hp['gamma']:.4g}, alpha={hp['alpha']:.4g}, "
 f"cv_MAE={hp['cv_mae']:.4f} ({time.time() - t0:.1f}s)")

curve = {"train_sizes": [], "mae_mean": [], "mae_min": [], "mae_max": [], "mae_all": []}
for n in TRAIN_SIZES:
 if n > len(pool_idx):
 continue
 maes = []
 for r in range(N_REPEATS_BY_SIZE[n]):
 sub_rng = np.random.RandomState(SEED * 1000 + r)
 train_idx = sub_rng.choice(pool_idx, size=n, replace=False)
 t0 = time.time()
 mae = fit_and_eval(X[train_idx], y[train_idx], X_test, y_test,
 gamma=hp["gamma"], alpha=hp["alpha"])
 maes.append(mae)
 print(f" N={n:6d} repeat={r} MAE={mae:.4f} ({time.time() - t0:.1f}s)")
 curve["train_sizes"].append(n)
 curve["mae_mean"].append(float(np.mean(maes)))
 curve["mae_min"].append(float(np.min(maes)))
 curve["mae_max"].append(float(np.max(maes)))
 curve["mae_all"].append([float(m) for m in maes])

curve["hyperparameters"] = hp
curve["n_test"] = int(n_test)
curve["target_std"] = float(np.std(y))
with open(RESULTS_DIR / "learning_curve.json", "w") as f:
 json.dump(curve, f, indent=2)
print(f"Wrote {RESULTS_DIR / 'learning_curve.json'}")

if __name__ == "__main__":
 main()

```